

Cybersecurity Incident Report

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: The client sent DNS queries over UDP to the DNS server at port 53 to resolve the domain name www.yummyrecipesforme.com.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "udp port 53 unreachable," indicating that the DNS server did not respond to the UDP request on port 53.

The port noted in the error message is used for: Port 53 is the standard port used for DNS (Domain Name System) services.

The most likely issue is: The DNS server is either not running the DNS service, the service is misconfigured, or network/firewall rules are blocking UDP traffic on port 53, causing DNS requests to fail.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: The incident was first reported at 13:24:32 (1:24 p.m.), as indicated by timestamps in the tcpdump log

Explain how the IT team became aware of the incident: Multiple customers reported being unable to access www.yummyrecipesforme.com and receiving the error message "destination port unreachable."

Explain the actions taken by the IT department to investigate the incident: The IT team replicated the issue by attempting to access the website, then used a network analyzer tool (tcpdump) to capture traffic and analyze DNS and ICMP packets

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): DNS requests were sent via UDP to port 53 on the DNS server IP 203.0.113.2. And ICMP error messages indicated that UDP port 53 was unreachable on the DNS server.

Note a likely cause of the incident: Most likely down due to a successful DOS attack or misconfiguration.